

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY
PROJECT PROPOSAL

PawLife

Web-Based Pet Care Management System

GROUP ID

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PREPARED BY

IT25100618 Sanvidu S.D.N

IT25101709 De Silva L.P.B

IT25102521 Subasinghe R.A.G.I

IT25103607 Balasooriya B.A.H.N

IT25101534 Alahakoon A.M.B.U

IT25103408 Warnakulasuriya G.A.D.T.S

1. Executive Summary

PawLife is a proposed web-based pet care management system designed to bring the clinical, operational, and administrative functions of a pet care centre onto a single, unified platform. The system will connect pet owners, veterinary staff, grooming teams, inventory personnel, and management through one accurate, real-time source of truth, replacing the paper based records, manual bookings, and disconnected departmental processes that currently limit visibility and slow down day to day operations.

This proposal outlines the background, objectives, scope, stakeholders, and the six core functional modules that make up the system, along with the student member responsible for delivering each module.

2. Background & Problem Statement

Pet care centres today typically rely on paper-based medical records, phone-in appointment bookings, and manually maintained inventory and grooming logs. This creates several recurring problems:

- Pet medical histories are scattered across paper files, making it difficult to retrieve accurate records during consultations.
- Manual booking processes lead to double bookings, missed appointments, and delayed communication with pet owners.

- Medicine and vaccine stock is tracked informally, increasing the risk of expired supplies and stock outs during clinical sessions.
- Grooming and specialty care services lack a structured way to record customer preferences, session notes, and service progress.
- Management has no real time, consolidated view of daily operations, staff workload, or business performance.

PawLife addresses these gaps by digitising and connecting every stage of the pet care journey from registration and booking, through clinical treatment and grooming, to inventory management and business reporting.

3. Project Objectives

- Centralise every pet's medical and service history into one secure digital record.
- Streamline scheduling across veterinary, grooming, and specialty care services through a unified booking system.
- Introduce real-time inventory tracking for vaccines and medical supplies, reducing waste and stock outs.
- Provide live operational visibility for centre managers, including staff workload and service status.
- Protect sensitive data through role-based access control and full audit logging.
- Deliver a convenient, self-service digital experience for pet owners.

4. Project Scope

The proposed system covers six core functional modules spanning clinical record-keeping, appointment management, inventory and supply chain control, grooming and specialty care workflows, operational and staff management, and system wide administration, security, and business intelligence. The scope includes design and development of the web application, the underlying database, and role-specific interfaces for each stakeholder group described in Section 5. Payment gateway integration, mobile native applications, and third party courier integrations are considered out of scope for the initial release and may be assessed for future phases.

5. Assumptions & Constraints

Assumptions

- Pet care centre staff have basic digital literacy and access to a computer or tablet at their workstation.
- A stable internet connection is available at the centre for real time booking, inventory, and dashboard updates.

- Existing paper-based medical and inventory records will be migrated into the system manually during rollout.
- The system is designed for a single pet care centre location at initial launch, with multi branch support treated as a future enhancement.
- Pet owners have access to a smartphone or computer with a web browser and, where applicable, SMS or email access for notifications.

Constraints

- The project is delivered within a 12 week academic development timeline by a team of six members.
- Each functional module is owned end to end by one team member, so module quality is dependent on individual delivery and requires close coordination to keep integration on schedule.
- No budget is allocated for paid third party services; the system relies on free-tier or open-source tools for hosting, SMS/email delivery, and testing.
- The system must run on standard modern web browsers without requiring a native mobile application for the initial release.

6. Stakeholder Overview

Requirement gathering was carried out across the key stakeholder roles who will use or depend on the system. Each role's responsibilities, expected system access level, and importance to the project are summarised below.

Stakeholder	Responsibilities	System Access	Priority
Business Owner	Growth, profitability, security and strategic analytics oversight	Full / Administrative	Critical
Centre Manager	Daily operations, staff workload and customer feedback review	Operational	Critical
Veterinary Officer	Diagnosis, vaccination records and prescriptions	Clinical	Critical
Customer Service Executive	Bookings, inquiries, reminders and billing receipts	Front-desk	High
Inventory Manager / Assistant	Vaccine and medicine stock, batch tracking and reorder alerts	Inventory	High
Grooming & Specialty Care Staff	Service customisation, status updates and care notes	Task-level	Medium
Pet Owners	Self-service booking, records, invoices and payments	Customer portal	Critical
Suppliers & Vendors	Supply of vaccines, medication and clinic stock	Inventory-only	Medium

7. System Architecture Overview

PawLife is organised into six functional modules, each owned end to end by one team member. Every module was scoped directly from stakeholder requirements and connects back to a shared data layer, ensuring that information entered in one module, such as a pet's medical record or grooming session is instantly reflected across the rest of the system.



	Functional Module	Assigned Member	Student ID
01	Digital Pet Profile & Health Record Module	De Silva L.P.B	IT25101709
02	Appointment Scheduling & Notification Module	Subasinghe R.A.G.I	IT25102521
03	Smart Inventory & Medical Supply Chain Management	Balasooriya B.A.H.N	IT25103607
04	Pet Grooming & Specialty Care Service Workflow	Warnakulasuriya G.A.D.T.S	IT25103408
05	Operational Centre & Staff Management Module	Alahakoon A.M.B.U	IT25101534

06	System Administration & Business Intelligence Module	Sanvidu S.D.N	IT25100618
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8. Functional Modules

The following section describes each module's purpose, key features, assigned owner, and the benefits it delivers.

MODULE 01

Digital Pet Profile & Health Record Module

Owned by De Silva L.P.B · IT25101709

A centralised platform maintaining every pet's complete medical history, from registration through to prescriptions, giving veterinary officers and pet owners a single accurate clinical record.

Key Features

- Register and maintain detailed pet profiles
- Record consultations, diagnoses, and lab reports
- Auto-calculate upcoming vaccination due dates
- Generate downloadable digital prescriptions

Expected Benefits

- Accurate, complete medical history
- Faster diagnosis during consultations
- Fewer missed vaccinations

MODULE 02

Appointment Scheduling & Notification Module

Owned by Subasinghe R.A.G.I · IT25102521

Manages veterinary, grooming, and specialty care bookings through one unified calendar, minimising scheduling conflicts and keeping pet owners informed.

Key Features

- Unified booking calendar across all services
- Automated SMS and email reminders
- Rescheduling and cancellation with live availability updates
- Logging of inquiries, complaints, and special requests

Expected Benefits

- Elimination of double bookings
- Higher appointment attendance
- Faster response to pet owners

MODULE 03

Smart Inventory & Medical Supply Chain Management

Owned by Balasooriya B.A.H.N · IT25103607

Provides real-time control over vaccine, medicine, and clinical supply stock, protecting the centre from expired stock, shortages, and untracked usage during treatment and grooming sessions.

Key Features

- Track vaccine and medicine stock levels with batch numbers and expiry dates
- Automated reorder alerts when supplies fall below minimum thresholds
- Deduct stock usage automatically during clinical and grooming sessions
- Generate waste and expired-stock audit reports for financial oversight

Expected Benefits

- Safer, always-valid clinical supplies
- Uninterrupted clinic operations
- Reduced financial losses from expired stock

MODULE 04

Pet Grooming & Specialty Care Service Workflow

Owned by Warnakulasuriya G.A.D.T.S · IT25103408

Supports the grooming and specialty care team in delivering, tracking, and documenting tailored services for each pet, with full visibility for pet owners.

Key Features

- Customise grooming service packages based on pet breed and coat condition
- Provide real time grooming progress updates to pet owners (e.g. Bathing, Styling, Ready)
- Log skin condition observations and behavioural notes during sessions
- Upload before and after photos for each grooming session

Expected Benefits

- Tailored, higher quality service delivery
- Greater transparency for pet owners
- Better informed veterinary follow up

MODULE 05

Operational Centre & Staff Management Module

Owned by Alahakoon A.M.B.U · IT25101534

Gives centre managers a real-time overview of daily operations, service progress, and staff performance to keep the clinic running smoothly.

Key Features

- Daily dashboard of appointments, boarding, and upcoming vaccinations
- Staff attendance, shift allocation, and workload tracking
- Live progress monitoring of ongoing services
- Alerts for missed appointments and overdue vaccinations

Expected Benefits

- Faster operational decisions
- Balanced staff workloads
- Fewer missed tasks

MODULE 06

System Administration & Business Intelligence Module

Owned by Sanvidu S.D.N · IT25100618

Ensures secure system operation while giving the business owner the analytics needed for data-driven decision-making.

Key Features

- Role-Based Access Control (RBAC) for every user type
- Full audit logs of user activity and system changes
- Customer trend and service popularity analysis
- Feedback collection to improve service quality

Expected Benefits

- Stronger data protection
- Full traceability of system activity
- Sharper, data-driven business strategy

9. Non-Functional Requirements

In addition to the functional modules described above, the system must satisfy the following quality attributes to be dependable in a live clinical and customer facing environment.

Category	Requirement
Security	All medical, personal, and financial data is protected with role-based access control, encrypted storage of sensitive fields, and full audit logging of user actions.
Performance	Core operations - search, booking confirmation, dashboard load - return results within 2 seconds under normal load.
Availability	The system targets 99% uptime during business hours, with scheduled maintenance windows communicated in advance.

Usability	Interfaces are role specific and require no more than basic digital literacy; critical workflows (booking, check-in, prescription issuance) are completed in three steps or fewer.
Scalability	The architecture supports growth in registered pets, staff accounts, and transaction volume without redesign, and allows future multi-branch expansion.
Reliability	Automated validation prevents double bookings, duplicate stock deductions, and conflicting cage/service assignments.
Data Privacy & Compliance	Pet owner and clinical data is handled in line with standard data protection principles, with access limited strictly to authorised roles.
Maintainability	Each module is built with clear ownership and documentation so that future team members can extend or modify functionality with minimal ramp-up time.

10. Implementation Timeline

Development is planned across four phases over a 12 week period, sequencing modules so that foundational data (authentication and pet health records) is available before dependent features such as scheduling, inventory, and grooming workflows are built on top of it.

Phase	Duration	Focus Area	Key Deliverables	Lead(s)
Phase 1	Weeks 1 - 3	Foundation & Health Records	Secure login and RBAC, digital pet profile registration, consultation and prescription records	De Silva L.P.B, Sanvidu S.D.N
Phase 2	Weeks 4 - 6	Scheduling & Inventory	Unified booking calendar, double-booking prevention, medicine stock tracking, automated reorder alerts	Subasinghe R.A.G.I, Balasooriya B.A.H.N
Phase 3	Weeks 7 - 9	Grooming & Specialty Care	Grooming package customisation, real-time service status updates, session notes and photo journal	Warnakulasuriya G.A.D.T.S
Phase 4	Weeks 10 - 12	Operations, BI & Final Integration	Operations dashboard, staff workload tracking, business intelligence and audit logging, end-to-end testing and deployment	Alahakoon A.M.B.U, Sanvidu S.D.N

11. Risks & Mitigation

The following risks have been identified as most likely to affect delivery or system quality, along with the corresponding mitigation approach for each.

Risk	Impact	Mitigation
Sensitive medical or financial data is exposed through a security gap	High	Enforce RBAC on every module, encrypt sensitive fields, and maintain full audit logs of data access and changes
Delays in one module affect integration of the overall system, since each module has a single owner	Medium	Define shared API contracts early, integrate modules incrementally each phase, and schedule regular cross-member check-ins
Inaccurate automatic stock deduction leads to shortages or overstocking	Medium	Validate stock transactions against session and appointment records, and reconcile with periodic audit reports
Low adoption of the system by clinic staff used to manual processes	Medium	Design intuitive, role-specific interfaces and provide short onboarding guidance for each user group
Scope creep beyond the six core modules within the academic timeline	Medium	Maintain a clearly defined module boundary per team member and treat additional features as future enhancements
Notification delivery (SMS/email) fails or is delayed	Low	Use reliable free-tier notification providers and log delivery status so failures can be identified and retried

12. Expected Benefits

Pet Owners

- Book and track services from anywhere
- Transparent, real time grooming and treatment updates
- Digital health records and invoices on demand

Veterinary & Grooming Staff

- Complete pet history available at every consultation or session
- Faster, paperless prescriptions and service documentation
- Automated vaccination and restock reminders

Management

- Real-time operational visibility across all services
- Consolidated inventory, staffing, and business analytics
- Secure, role-based access to sensitive data

13. Conclusion

PawLife brings together clinical care, scheduling, inventory management, grooming services, operations, and business intelligence into one connected platform. By replacing manual, siloed processes with a unified system, the project aims to improve efficiency, strengthen data accuracy, and deliver a significantly better experience for pet owners, staff, and management alike. The modular structure outlined in this proposal, with clear ownership assigned to each team member and a phased delivery plan with identified risks and mitigations, provides a practical, low-risk foundation for the system's design and development.